

Exp No: 18 GENERATION OF PWM USING TIMER INTERRUPT

PROGRAM:

```
#include <reg51.h>

sbit PWM = P1^0;

unsigned char count = 0;

unsigned char duty = 100; // Change this value (0–200)

void timer0_ISR(void) interrupt 1
{
    TH0 = 0xFC; // Reload for 1ms delay
    TLO = 0x66;

    count++;

    if(count < duty)
        PWM = 1; // ON time
    else
        PWM = 0; // OFF time

    if(count >= 200) // Total period = 200ms
        count = 0;
}

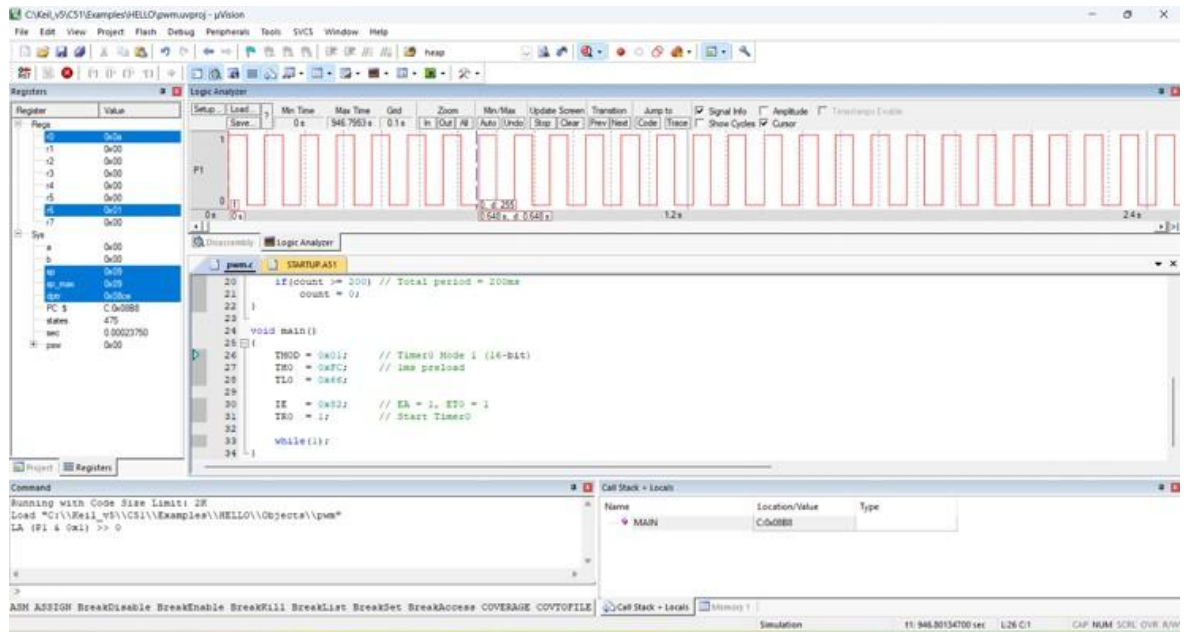
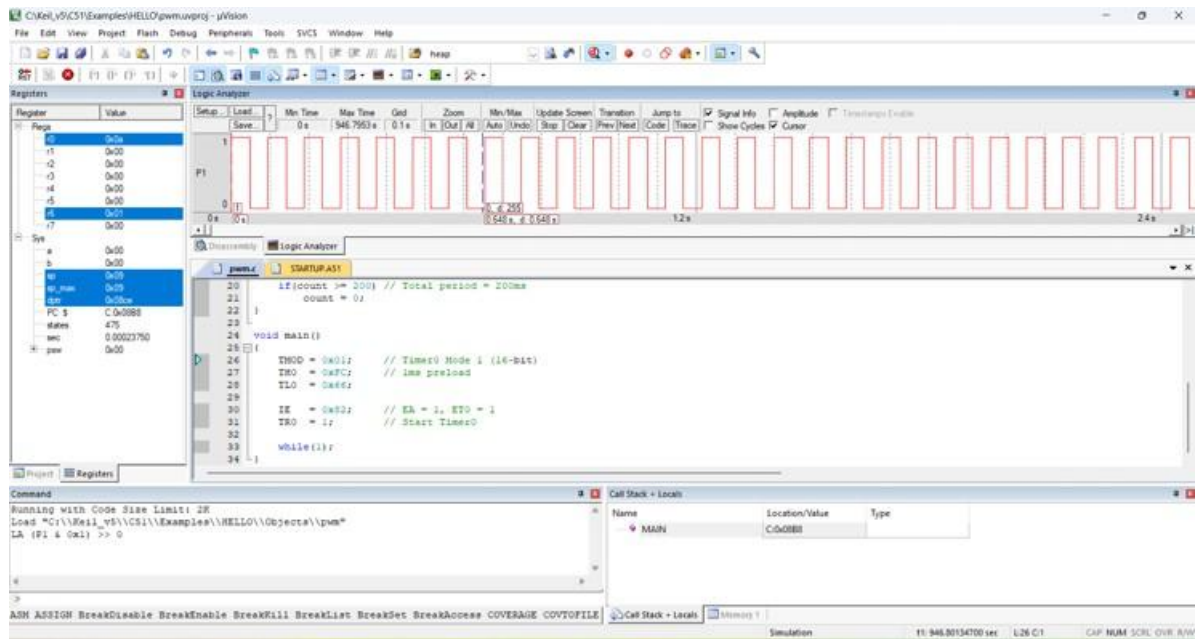
void main()
{
    TMOD = 0x01; // Timer0 Mode 1 (16-bit)
    TH0 = 0xFC; // 1ms preload
    TLO = 0x66;

    IE = 0x82; // EA = 1, ET0 = 1

    TR0 = 1; // Start Timer0

    while(1);
}
```

Output



C:\Keil_v5\CS1\Examples\HELLO\gwmuvproj - µVision
 File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers
 Register Value
 R0 0x00
 R1 0x00
 R2 0x00
 R3 0x00
 R4 0x00
 R5 0x00
 R6 0x01
 R7 0x00
 Sys 0x00
 SPC 0x00
 SP 0x00
 PC 0x0088
 Status 475
 sec 0.00023750
 usec 0x00

Logic Analyzer
 Setup Load Save Min Time Max Time Gnd Zoom Min/Max Update Screen Transition Jump to Signal Info Amplitude Timebase Enable
 0s 0s 346.763s 0.1s In Out All Auto Undo Stop Clear Prev Next Code Trace Show Cycles Cursor
 P1
 0 1
 0.4s 0.8s 1.2s 2.4s
 0.46s, 8.056s

Disassembly
 20 if(count >= 200) // Total period = 200ms
 21 count = 0;
 22
 23
 24 void main()
 25 {
 26 THOD = 0x01; // Timer0 Mode 1 (16-bit)
 27 THO = 0x0C; // 1ms preload
 28 TLO = 0x00;
 29
 30 IE = 0x02; // EA = 1, ET0 = 1
 31 TR0 = 1; // Start Timer0
 32
 33 while(1);
 34 }

Command
 Running with Code Size Limit: 2K
 Load "C:\Keil_v5\CS1\Examples\HELLO\Objects\gwm*"

Call Stack - Local
 Name Location/Value Type
 MAIN C:0088

Call Stack - Local Memory
 Simulation 11.946.00154700 sec L26 C1 CAP NUM SCRL OVR R/W